Operating manual

Forcebit sensor

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Contents

1. Safety warnings 4
Definition of warnings 4
General warnings 4
2. Conventional usage 6
3. Technical specifications, assembly & maintenance 6
Important precautions during installation 6
Protection against external factors 6
Mounting/installation 6
Recharging the system 7
Maintenance and storage 7
4. Waste disposal 7
5. Transport and storage 8

Disclaimer

While this manual has been prepared very diligently, potential error sources only truly reveal itself when it is used at the customers premises. Therefore, we deeply appreciate any feedback that can help us improve our service and the safety of our product.

1. Safety warnings

Definition of warnings

DANGER!

Indication of a possible dangerous situation. Ignoring the security measures may cause death or serious injuries.

WARNING!

Indication of a possible dangerous situation. Ignoring the security measures may cause pain or injuries.

CAUTION!

Indication of a risk for damaging surrounding property if the corresponding protective measures are discarded.

General warnings

This device may only be installed by skilled personnel, who is qualified to evaluate the potential risks. The complete manual has to be read and understood prior to handling the device. The personnel has to be briefed on the general safety measures on the plant regarding the machine in which the sensor will be installed. On no account, this manual shall overrule existing safety measures!

Deviations of any kind cannot lead to any claim for incurred losses from the manufacturer. The customer is not allowed to make any modifications to the product as that will invalidate the warranty.

DANGER!

Incorrect installation may lead to unwanted machine vibrations and possible loosening/flying off of the sensor from the attached shaft. Possible debris may lead to cascaded damages to the machines and it's functional behavior. The consequences of such failure needs to be evaluated on a case-by-case bases and correct measures should be taken to avoid injuries. The customer understands this risk and takes appropriate measures prior to installing the sensor.

DANGER!

All loose equipment or components has to be removed from the machine after installation! Any parts such as screws, wrenches etc. present should stay at a safe distance for the rotating parts to avoid being catapulted.

DANGER!

Unintentional startup of the system may cause parts to move/rotate causing serious injury by any personnel in the direct vicinity. During all mounting, dismounting, repair and maintenance, the system must be powered off.

DANGER!

A Forcebit sensor contains up to three integrated rechargeable lithium-ion coin cells (VARTA CoinPower CP1654, 3.7 V, approx. 0.22 Wh per cell). Improper handling of lithium-ion batteries may lead to fire, explosion, or burns. Do not short-circuit, puncture, crush, heat above 75 °C, or incinerate the batteries. Do not open the device or attempt to replace the batteries yourself. In case of visible damage, swelling, leakage, or abnormal heating, discontinue use immediately and contact Forcebit BV.

WARNING!

The sensor will become warm during operation, allow a cool down period prior to touching the shaft to avoid burn injury.

WARNING!

The internal circuitry in the gateway will become warm during operation, opening the casing and touching the PCB might cause burn injury.

2. Conventional usage

The sensor is used for measuring dynamic quantities on fast moving systems. It provides contact-free data transfer. The product has to be used explicitly for the purpose that it was ordered.

3. Technical specifications, assembly & maintenance

The technical specifications are as presented on the website www.forcebit.eu

Specific battery information

The Forcebit sensor is equipped with up to 3 integrated rechargeable lithium-ion coin cells of type VARTA CoinPower CP1654.

- Nominal voltage: 3.7 V
- Typical capacity: approx. 60 mAh
- Energy content per battery: approx. 0.22 Wh
- Total energy content per sensor: max. 0.66Wh

The batteries are permanently integrated into the device and are not user-replaceable. Opening the device or replacing the batteries by the user is not permitted and will invalidate the warranty.

Important precautions during installation

Forcebit sensors are precision measurement devices and must be handled with extreme care. Knocking, deforming, dropping the device might cause unexpected plastic deformation thus permanently damaging the sensor. If not in use, the sensor should be stored away carefully.

Protection against external factors

Unless in open air, Forcebit sensors should always be mounted with its delivered cover and is, therefore, insensitive to moisture, dust, dirt and other foreign matter.

Mounting/installation

The mounting procedure consists out of several steps.

DANGER!

The sensor cannot be used outside its operating limits. Exceeding the temperature limits until 75°C will only reduce performance but heating it up to 90°C might cause breakdown of the batteries. Exceeding the maximum velocity might cause induced vibrations or breakdown of the sensor entirely.

Step 1: Properly clean the surface with a degreaser

The surface area that will be used by the sensor has to be appropriately degreased. Any solvent-based degreaser would be sufficient. It is important that the final surface is clean and dry. Additional surface preparation such as sanding is not necessary unless the surface is rough > 3.2 µm.

Step 2: Mount the sensor with the M3 screw/glue/tape until the sensor is firmly fixed

For the ACCbit & TELbit

- The sensor should be mounted with an M3 screw and firmly attached until there is no more relative motion between the sensor and the shaft. Mind that the sensor is attached to a machine and that the standard safety regulations concerning “EU Machinery Directive 2006/42/EC”.

For the DOT

- The sensor can be attached with a double-sided tape, epoxy glue or magnets. In the case of double-sided tape or magnets, the user must verify that the sensor is not subjected to vibrations outside of the sensor range to avoid detachment.
- For the use if epoxy glue attachment, please for the respective guidelines listed on the glue documentation. Mind that potential dis-attachments can happen and the users should take appropriate measurements to avoid injury or damage when the sensors would fly off.

Mind that the sensor is attached to a machine and that the standard safety regulations concerning “EU Machinery Directive 2006/42/EC”.

Recharging the system

To recharge the sensor, ensure that the machine is completely stationary and sufficiently cooled down to avoid any accidental burns. The temperature while charging should stay between 15°C and 45°C. Connect the USB C port to the USB cable and wait maximally 2 hours until complete recharge.

Maintenance and storage

When the sensor is removed, it should be cleaned with degreaser and stored at room temperature. The user is not allowed to make custom repairs or adjustments but has to send it back to Forcebit for repair and maintenance.

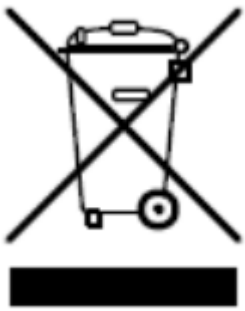
If contamination is suspected during operation, remove the protective cover and inspect the sensor cavities for debris, dirt, or fluids. If contamination is found, remove the sensor, clean it, and reinstall it according to the installation instructions.

Improper storage at elevated temperature (45°C), the batteries will self-discharge. Increasing the temperature will further increase the rate of discharge.

4. Waste disposal

All electrical and electronic products must be disposed of as hazardous waste. The correct disposal of old equipment prevents ecological damage and health hazards.

Statutory waste disposal mark



The electrical and electronic devices that bear this symbol are subject to the European waste electrical and electronic equipment directive 2002/96/EC. The symbol indicates that, in accordance with national and local environmental protection and material recovery and recycling regulations, old devices that can no longer be used must be disposed of separately and not with normal household garbage. As waste disposal regulations may differ from country to country, we ask that you contact your supplier to determine what type of disposal or recycling is legally applicable in your country. The integrated lithium-ion battery must not be disposed of with household waste.

5. Transport and storage

The Forcebit sensor contains up to three rechargeable lithium-ion coin cells (VARTA CoinPower CP1654, total energy content ≤ 0.66 Wh) integrated into the device.

The product may be transported and shipped with the batteries installed, provided that:

- the device is not damaged,
- the batteries show no signs of swelling, leakage, or overheating,
- the sensor is protected against short-circuit and mechanical damage,
- transport and storage temperature remains between –20 °C and +45 °C.

The product qualifies as “lithium-ion batteries contained in equipment” and does not require special dangerous goods documentation for road, sea, or air transport when shipped in compliance with applicable regulations.